

MinJun Chang

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Education

Georgia Institute of Technology , PhD in Civil Engineering	Aug 2025 - Current
• Robotics and Intelligent Construction Automation Laboratory	
Yonsei University , BS in Mechanical Engineering	Mar 2019 - Feb 2025
• Coursework: Mechatronics, Mechanism Design, Mechanical Element Design, Dynamics, Numerical Analysis	

Research

RICAL Group , Georgia Institute of Technology – Atlanta, GA, USA	Aug 2025 – Current
Graduate Research Assistant, Supervisor: Dr. Yong K. Cho	
• Deep-Learning based real-time human motion recognition with localization with IMU sensors in construction	
• Whole body control of a humanoid for dexterous tasks in construction and manufacture	
Dynamic Robotic System Laboratory , Seoul National University – Seoul, KR	Jul 2024 – Dec 2024
Undergraduate Research Assistant, Supervisor: Dr. Jaeheung Park	
• Model-free reinforcement learning framework with a state estimator neural-network for bipedal locomotion	
Machine Learning and Control System Laboratory , Yonsei University – Seoul, KR	Jul 2022 – Jan 2023
Undergraduate Research Assistant, Supervisor: Dr. Jongeun Choi	
• Developed an enhanced localization algorithm with control input delay compensation	
Mechanobiology and Soft Materials Laboratory , Yonsei University – Seoul, KR	Jul 2020 – Jun 2021
Undergraduate Research Assistant, Supervisor: Dr. Hyungseok Lee	
• Participated in development of portable Standing Surface Acoustic Wave cell alignment device on transparent bridge lumen structure	

Publications

Real-Time Temporal Convolution based Worker Motion Recognition with Three IMU Sensor Combinations , 43rd International Symposium on Automation and Robotics in Construction, <i>Review</i>	Jun 2026
<i>MinJun Chang</i> , Sai Machiraju, Francis Baek, Yong K. Cho†	
Robust Symmetric Bipedal Locomotion Development via Simultaneous State Estimator Neural Network Training , The 20th Korea Robotics Society Annual Conference, Poster	Feb 2025
<i>MinJun Chang</i> , Jaeyong Shin, Jaeheung Park†	
State prediction-based control input delay compensation for autonomous driving systems , The 18th Korea Robotics Society Annual Conference, Oral	Feb 2023
<i>MinJun Chang</i> , H.W. Nam, S.Y. Choi, J.H. Yang, J.H. Yang, Jongeun Choi†	

Patents and Copyrights

Autonomous Driving Auxillary Cart Robot for Manufacture	Aug 2024
KR10-2024-0177135, under prosecution	
The Urine Examination Apparatus and Controlling Method of the Same	Dec 2020
KR10-2020-0176792, under prosecution	
Eye Tuner: Media Literacy Program based on Pupil Tracking by Computer Vision	Nov 2021
Korea Copyright Commission, C-2024-039138, Registered-Active	

Professional Experience

Hanwha Systems(Space&Defense) , Satellite Software Engineer	Jan 2025 – May 2025
• ARMv7 MCU internal communication software programming for Small Synthetic Aperture Radar (SAR) Satellite	
GOLE Robotics , Robotics Engineer	Apr 2024 – Jun 2024
• Implemented global and local robot path planning algorithm on ROS2 for construction delivery robot WERO • Developed actuator controller package with C++/Python binding enabling python API usage of C++ source	
DRIMAES , Embedded Software Engineer	Oct 2022 – Mar 2024
• Linux, ARM MCU software/firmware programming for various products • Developed various communication protocols (Serial, MQTT, REST, CAN) • Implemented multiple virtual container management technique on cross-platform systems	
SIOT Infotech , Embedded Software Engineer	Feb 2022 – Oct 2024
• Developed embedded software for Mediatek, ESP chipsets and enhanced custom OpenWRT OS kernel	

Selected Awards

1st place, National ICT Smart Device Competition , Korean Ministry of Science and ICT	Aug 2024
• Awarded by the Minister of Science and ICT of Republic of Korea • Led a team of 5 in developing an Autonomous Manufacture Assistant CARTRASCHE	
1st place, 2022 Autonomous Driving Robot Racing Contest , Korean Robotics Society (KRoS)	Nov 2022
• Participated as Localization team member • Developed a control algorithm utilizing LiDAR, IMU, and GPS sensor fusion for collision avoidance	
Selection, Hanium Contest , Federation of Korea Information Industries	Nov 2021
• Led a team of 4 in developing Personalized Content Literacy program EYE-TUNER • Implemented pupil tracking algorithm for the program	
2nd place, Medical Hack 2021 , Busan City	Nov 2021
• Awarded by the mayor of Busan City • Implemented posture prediction algorithm with multiple load-cell sensors	
2nd place, Yonsei IHEI Workstation , Yonsei University	Jul 2020

Projects

Awaresite : Intelligent PPU System for Construction Productivity and Safety	Aug 2025 - Current
• Deep-Learning based worker motion recognition using IMU data • Zone scale indoor localization using BLE beacons	
CARTRASCHE: Autonomous Driving Auxiliary Cart Robot	Mar 2023 - Jul 2024
• Developed autonomous driving mobile robot with rotating shelf system using SLAM for navigation in ROS • Implemented custom RC filter and encoder-less motor control algorithm for activation • Managed the project flow and system overall management • Tools Used: C++, Python, ROS, LinuxOS	
UAV Fleet Management System for Robot Cluster in Factory	Aug 2023 - Dec 2023
• Implemented task scheduling and allocation algorithm based on order status for multi-robot network • Tools Used: Python, ROS, MQTT	
FennecBot: Industrial Anomaly Detection Mobile Robot	May 2023 - Aug 2023
• Developed multi-modal deep learning network for pipeline anomaly detection and the classification of pipeline leakage using RGB camera, and ultrasonic/acoustic sound camera • Operated on Scout mini with line-tracing algorithm detecting pipe leakage within Hyundai HI. factory	

Honors

Federico Stubbe Graduate Student Fellowship , Georgia Institute of Technology	Aug 2025
Fellowship granted to competent graduate students	
Next Generation Engineer , Institute for Promotion of Engineering and Science of Korea	Dec 2024
Honor society for outstanding young engineers in Republic of Korea	

Technologies

Programming: Python, C++, C, JavaScript, MATLAB

Frameworks/Tools: ROS, IsaacSim/Lab, PyTorch, Docker, AWS, FastAPI, Solidworks, Fusion360

Hardware: ARM V7, Jetson Xavier, Jetson Nano, RaspberryPi, Arduino, STM32, ESP32, Unitree Go1

Languages: Korean(Native), English(Fluent), Japanese(Median)